Dual-graph Augmented Contrastive Learning for Imbalanced Multi-modal Representation

Shuang Yao[a], Wenfeng Han[a], Xinxin Wei[a], Yao Dong[a,b,c,*], Tianyi Zhang[a], Yongfeng Dong[a,b,c]

[a] *School of Artificial Intelligence, Hebei University of Technology, Tianjin, 300401, China*
[b] *Hebei Province Key Laboratory of Big Data Computing, Tianjin, 300401, China*
[c] *Hebei Engineering Research Center of Data-Driven Industrial Intelligent, Tianjin, 300401, China*

Abstract

Data imbalance poses a significant hurdle to multi-modal representation and integration. While existing imbalance techniques in multi-modal learning have demonstrated remarkable performance, their focus has primarily been on addressing sample quantity imbalance, often overlooking cross-modal structural heterogeneity and representation conflicts. This underscores the urgency for research that holistically tackles both sample imbalance and modality-specific topology challenges. Furthermore, most existing imbalanced multi-modal models fail to dynamically weight modality importance and struggle with incomplete cross-modal information fusion. To address these issues, we propose Dual-graph Augmented Contrastive Learning for Imbalanced Multi-modal Representation Fusion (DACL). Our framework introduces a dual augmentation strategy: (1) Modality-specific topology augmentation based on structural differentiation, and (2) Feature augmentation leveraging modality significance. We then integrate cross-modal contrastive learning with adaptive modality fusion to explore inter-modal relationships and weight each modality dynamically. Extensive experiments on diverse multi-modal datasets confirm our method's superior effectiveness.

Keywords: Imbalanced multi-modal data; Multi-modal representation; Multi-modal integration; Multi-modal graph contrastive learning; Dual augmentation

1. Introduction

Data imbalance remains a critical challenge in multi-modal representation learning and cross-modal integration, often leading to biased models that prioritize majority modalities or classes while neglecting underrepresented ones. In complex multi-modal systems where data originates from diverse sources such as visual, textual, and auditory inputs, the inherent heterogeneity of feature spaces and sampling frequencies across modalities exacerbates this imbalance problem. Representation learning in such scenarios must not only capture the complementary and redundant information across modalities but also address the fundamental disparities in data distribution and feature richness. Cross-modal integration further compounds these challenges, as conventional fusion approaches tend to be dominated by the majority modality features, resulting

*Corresponding author.
Email address: dongyao@hebut.edu.cn (Yao Dong)

in suboptimal performance for minority classes and modalities. This imbalance-induced bias becomes particularly problematic in real-world applications like medical diagnosis (where clinical notes may outweigh medical imaging samples)[1, 2] or multimedia analysis (where visual data often dominates textual annotations)[3], ultimately limiting the model's ability to learn robust, generalizable representations that fairly represent all modalities and classes.

Previous imbalance issues usually existed in unimodal settings. Unimodal graph tasks refer to graph-based learning problems processing single-modality data, where graph structures capture inherent relationships within the data domain [4]. Researchers have proposed two primary approaches: data-level methods and algorithm-level methods. Data-level methods encompass techniques such as data interpolation [5, 6, 7, 8], adversarial generation [9, 10], and pseudo-labeling [11]. These methods aim to adjust the original data distribution to achieve a balanced class distribution. Algorithm-level approaches, on the other hand, focus on adapting learning algorithms to modify the training process of the model, with the goal of mitigating challenges arising from imbalanced class distribution. These approaches include model refinement [12, 13], loss function engineering [14, 15, 16], and post-hoc adjustments [17, 18]. The current works have been achieved in unimodal graph tasks, but they usually don't take into account both class quantity imbalance[19] and topology imbalance[20] together. Class quantity imbalance involves the uneven distribution of training samples across different classes, while topology imbalance stems from structural disparities where nodes exhibit vastly different connectivity patterns. Both issues considerably impact model performance: class imbalance creates biased feature spaces dominated by majority class gradients, whereas topology imbalance leads to unequal information propagation between well-connected and sparsely-connected nodes. In multi-modal contexts, these problems intensify as different modalities often display distinct imbalance characteristics, further complicating cross-modal feature integration. Furthermore, the existing methods that address the imbalanced uni-modal issues cannot be directly extended to multi-modal scenarios due to inherent modality heterogeneity and distinct feature spaces.

Consequently, how to simultaneously and effectively address class quantity and topology imbalances in multi-modal data is a crucial challenge for us. Current multi-modal approaches also typically assign equal weights to all modalities, failing to account for their varying relevance across different samples and classification scenarios. In addition to these challenges, the extraction of multi-modal information is another crucial area that warrants further exploration. During the process of feature extraction, previous methods often overlook the dynamic modality importance during integration and may fall short in capturing complex inter-modal dependencies.

To tackle these limitations, we propose a novel framework: Dual-graph Augmented Contrastive Learning for Imbalanced Multi-modal Representation (DACL). The dual graph augmentation module (DGA) includes topology augmentation based on degree differentiation and feature augmentation based on attribute significance. To rectify the topology imbalance, we employ ego-centric network interpolation and similarity-based sampling techniques to add and drop edges. On the feature level, we leverage centrality measures to identify essential feature dimensions and introduce corruption to less important feature dimensions through noise injection. The multi-modal graph contrastive learning module (MmGCL) attempts to capture complex inter-modal dependencies from multi-modal data. In contrast, the multi-modal integration module (MmI) is intended to facilitate adaptive cross-modal fusion among the modalities. The main contributions of our work are as follows:

- We propose a novel dual graph augmentation method to address the challenge of topology imbalance in multi-modal data. This method utilizes topology-level interpolation and sampling to expand and refine node neighborhoods. Additionally, we employ feature-level masking to reduce the impact of less significant feature dimensions.

- We design a joint contrastive loss function that combines both contrastive loss and class-balanced loss. This integrating loss function is employed to optimize the multi-modal graph contrastive learning encoder, aiming to achieve a robust feature representation across different modalities.
- We introduce a modality-level fusion self-attention mechanism designed to capture dynamic modality importance and facilitate their collective contribution to multi-modal integration and downstream tasks.
- The proposed framework is leveraged to conduct multi-modal imbalanced node classification tasks and achieves superior performance compared with other state-of-the-art baseline algorithms.

The remainder of this paper is organized as follows: Related works are reviewed in Section 2, followed by the proposed method in Section 3. Experimental results and analysis are detailed in Section 4, with conclusions discussed in Section 5.

2. Related Works

As data science and machine learning evolve, dealing with complex multi-modal data becomes challenging. Traditional methods often struggle with imbalanced multi-modal data, resulting in poor performance. Therefore, this section focuses on three areas closely related to our research: imbalance learning, multi-modal learning, and graph contrastive learning.

2.1. Imbalance Learning

In real-world applications, imbalanced classification is common, where one class has significantly more samples than others, influencing model performance. Existing methods address class quantity imbalance through data- and algorithm-level algorithms. Data-level algorithms include data interpolation, adversarial generation, and pseudo-labeling. Zhao et al.[6] introduced GraphSMOTE for graphs, synthesizing minority nodes by interpolating between genuine minority nodes in the embedding space. Duan et al.[10] presented SORAG, employing adversarial generation for synthetic minority nodes, using GAN for unlabeled and cGAN for labeled nodes. Zhou et al.[11] utilized GNN for pseudo-labeling unlabeled nodes, identifying reliable ones and dynamically enriching training nodes via reinforcement learning. Algorithm-level techniques include model refinement, loss function engineering, and post-hoc adjustments. Wang et al.[13] proposed EGCN, refining models with adjusted aggregation operations locally and emphasizing minority classes globally based on imbalance ratios. Cui et al.[14] assigned higher importance to loss from minority class samples during training, enhancing model responsiveness to errors. Yun et al.[18] employed class prototype-based inference, adjusting predictions post-training during inference.

Nevertheless, the direct application of these methods to graph scenarios fails to address the unique topology imbalance specific to graphs. Chen et al.[21] proposed ReNode to adjust node influence based on their positions relative to class boundaries, but homophily assumptions limit it. Sun et al.[22] introduced metrics for topology imbalance, focusing on under-reaching and over-squashing perspectives, but PASTEL heavily relies on labeled data, potentially limiting its practical use. In addition, most of them only concentrated on either class quantity imbalance or topology imbalance, both of which influenced the model's performance considerably.

To this end, we propose the DACL method to address topology imbalance by augmenting graphs at both feature and topology levels. Furthermore, we optimize a multi-modal graph contrastive learning encoder with graph contrastive loss and class-balanced loss to acquire balanced feature representations, resolving class imbalance.

2.2. Multi-modal Learning

With the rapid development of multimedia, data exists in more complex and diverse forms, contributing to the popularity of multi-modal learning. Using local manifold learning, Nie et al.[23] proposed a multi-modal learning model with adaptive neighbors to learn a consensus affinity graph from multi-modal data. In addition, Nie et al.[24] further presented an auto-weight graph-oriented method for automatically allocating an appropriate weight for each modality. To fully exploit the effectiveness of GCN, Li et al.[25] proposed a multi-modal semi-supervised learning model that exploits the graph information from the multiple modalities adaptively using combined Laplacians. Chen et al.[26] proposed a joint deep learning model by learning an underlying feature representation from heterogeneous modalities and using the DSA function to achieve graph fusion. Wu et al.[27] integrated the reconstruction error and Laplacian embedding to explore the original space from both feature and topology perspectives while also proposing an orthogonal normalization method to address the issue of orthogonal constraints.

Existing multi-modal learning methods often overlook data imbalance, prioritizing modality combination efficiency over feature interaction complexity. Our research tackles this imbalance with a dual graph augmentation module. We optimize the graph contrastive learning encoder with a joint loss function to unveil consistently balanced representations across modalities. Using a multi-modal fusion self-attention mechanism, we discern modality importance and integrate information for a comprehensive underlying representation[28, 29].

2.3. Graph Contrastive Learning

Graph Contrastive Learning is popular in node representation learning. By contrasting positive and negative data, contrastive learning produces discriminative representations. Sun et al.[30] presented a novel approach to treating the graph structure of different scales (e.g., nodes, edges, and triangles) as contrastive modality graphs based on local-global mutual information maximization. Then, Hassani et al.[31] suggested performing node diffusion and contrasting node representations with augmented graph representations to learn both node-level and graph-level representations. Zhu et al.[32] proposed a joint, adaptive data augmentation technique to learn node representations using other nodes as negative samples at both topology and node attribute levels. Jovanovic et al.[33] introduced a new method that increases the adversarial robustness of the learned representations through adversarial transformations and random graph augmentation. Zhang et al.[34] proposed an effective personalized augmentation framework to learn graph representation, the first to assign personalized augmentation types to each graph based on its distinctive characteristics. Zhuo et al.[35] used the updated augmented modality graphs and the original graph to jointly learn node representations by optimizing an efficient channel-level contrastive objective.

Existing graph contrastive learning methods excel in node representation but neglect topology imbalance. Our framework enhances key feature representations by measuring feature centrality in the original graph. We adjust imbalanced topology through interpolation and sampling, yielding balanced augmented graphs. Joint optimization of class balance loss and graph contrastive loss maximizes consistency between original and augmented graphs, capturing balanced feature representations and enhancing model performance.

3. The Proposed Method

The DACL first enhances the data at feature and topology levels through the dual graph augmentation module. For feature augmentation, we devise a method based on attribute significance to retain essential node features. For topology augmentation, we design a strategy based on node degree differentiation to adjust the imbalanced topology structure of the modalities. Subsequently, we construct the multi-modal graph contrastive learning module, which combines graph contrastive loss and class-balanced loss to optimize the graph contrastive learning encoder for better extraction of cross-modal information. Finally, through the multi-modal integration module, we design a multi-modal fusion self-attention mechanism to emphasize the distinct contributions of different modalities, thereby enhancing the model's ability to integrate information from diverse modalities. The architecture of DACL is illustrated in Figure 1.

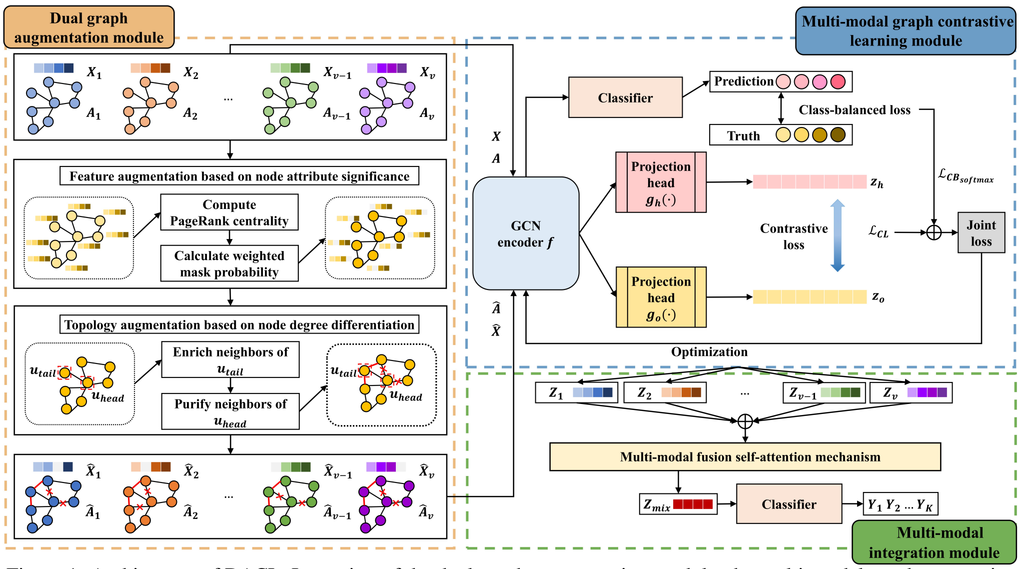


Figure 1: Architecture of DACL. It consists of the dual graph augmentation module, the multi-modal graph contrastive learning module and the multi-modal integration module. X and $\hat{X}$ represent the feature matrix of the original and augmented modality graphs. In contrast, A and $\hat{A}$ are the adjacency matrix of the original and augmented modality graphs, respectively. Z_v is the feature representation of the extracted v-th modality. The projection $g(\cdot)$ is a two-layer multilayer perceptron.

3.1. Preliminaries

Denote the given multi-modal data as $\mathcal{X} = X_1, X_2, \cdots, X_v$ to construct the modal representation $\mathcal{G} = G_1, G_2, \cdots, G_v$ respectively, where X_i is the data from i-th modality with totally v modalities. Let $G = (U, \mathcal{E})$ be a graph, where $U = u_1, u_2, \cdots, u_N$ is the set of N nodes, $\mathcal{E} \subseteq U \times U$ represents the edge set. We denote $X_i = [x_1, x_2, \cdots, x_N] \in \mathbb{R}^{N \times B_i}$ and $A \in \{0, 1\}^{N \times N}$ as the feature matrix and the adjacency matrix of the i-th modality respectively, where B_i is the feature dimension from i-th modality, x_j is the feature vector of node u_j and $A_{ij} = 1$ if u_i,$u_j \in E$. Each node belongs to one of K latent communities $D_1, D_2, \cdots, D_K$.

3.2. Dual graph augmentation module

3.2.1. Feature Augmentation based on Attribute Significance

We generate graph modals by performing feature augmentation based on attribute significance. In feature augmentation using attribute significance, we employ PageRank centrality as a metric function to gauge the significance of nodes. PageRank centrality relies on the weights calculated by the PageRank algorithm [36], which determines the importance of nodes within the network. The algorithm propagates influence along directed edges, considering certain nodes as virtual nodes accumulating the highest influence. The formal definition of the centrality values is as follows:

$$\sigma = \alpha A D^{(-1)} \sigma + 1, \tag{1}$$

where $\sigma \in R^N$ is the vector of PageRank centrality scores for each node, and α is a damping factor.

Our PageRank application differs from traditional node ranking. Instead of ranking nodes by importance, we use these centrality scores as weights to determine feature dimension significance in Equations (2) and (3).

We presume that feature dimensions often associated with prominent nodes are likely significant. As a result, we define the weights of feature dimensions as follows. When dealing with sparse one-hot node features x_q, we calculate the weight of feature dimension i as:

$$w_i^f = \sum_{q \in U} x_{qi} \cdot \varphi_c(q), \tag{2}$$

where $\varphi_c(\cdot)$ is a node centrality measure used to quantify node importance, i.e., PageRank centrality. $x_{qi} \in 0, 1$ indicates the occurrence of dimension i in node q, and $\varphi_i(q)$ measures the node importance of each occurrence.

For dense, continuous node features x_o of node o, where x_{oi} denotes feature value at dimension i, the occurrence of each one-hot encoded value cannot be easily counted. Then, we turn to measure the magnitude of the feature value at dimension i of node o by its absolute value $|x_{oi}|$. Formally, we calculate the weights as:

$$w_i^f = \sum_{o \in U} |x_{oi}| \cdot \varphi_c(o). \tag{3}$$

To obtain the probability linked to feature importance, we proceed to normalize the weights. This is done formally as follows:

$$p_i^f = \min\left(\frac{s_{max}^f - s_i^f}{s_{max}^f - \mu_s^f} \cdot p_{fdrop}, p_\tau\right), \tag{4}$$

where $s_i^f = log w_i^f$, s_{max}^f and μ_s^f is the maximum and the average value of s_i^f respectively. p_{fdrop} is a hyperparameter that controls the overall magnitude of feature augmentation, and p_τ is the given feature importance threshold. We randomly sample a mask vector $m \in \{0, 1\}^B$ to hide a portion of the dimensions in the node feature. Each element in the mask m is sampled from a Bernoulli distribution, i.e., $m_i \sim Ber\left(1 - p_i^f\right)$, where the hyperparameter p_i^f indicates how significant the i-th dimension of node characteristics is. Thus, the augmented node feature $\hat{X}$ is computed as:

$$\hat{X} = [x_1 \circ m, x_2 \circ m, \cdots, x_N \circ m]. \tag{5}$$

3.2.2. Topology Augmentation based on Degree Differentiation

After undergoing attribute importance-based feature augmentation, we employ topology augmentation based on node degree differentiation to generate the final augmented modality graph. First, we set a threshold ζ to distinguish head nodes and tail nodes. Then, based on the cosine similarity of their representations, we construct the similarity matrix S between node pairs, which is computed as:

$$S_{i,j} = \mathrm{sim}(h_i, h_j) = \begin{cases} \frac{1}{\tau} \cdot \frac{h_i^\top h_j}{\|h_i\| \cdot \|h_j\|}, & i \neq j \\ 0, & i = j \end{cases} \tag{6}$$

where h_i represents the node representation of node u_i, $\|h_i\|$ and $\|h_j\|$ are the norms of the vector, τ is an adjustable parameter, h_i^T denotes the transpose matrix of h_i. For any tail node u_{tail} in each modality, we sample a node u_{sam} from the multi-modal distribution $Multi\,(s_{tail})$, where s_{tail} is the row vector of S corresponding to u_{tail}. Then, we generate a new similarity-aware neighborhood for u_{tail} by interpolating between neighbor distributions of u_{tail} and u_{tar}. Here, the neighbor distribution for node u is defined as:

$$p(q|u) = \begin{cases} 1/|N(u)|, & q \notin N(u) \\ 0, & q \in N(u) \end{cases}, \tag{7}$$

where p is the conditional probability. Node similarity is utilized as the interpolation ratio to avoid deleterious connections. The neighbor distribution u_{tail} is computed as:

$$p_{sam}\,(q|u_{tail}) = \phi p\,(q|u_{tail}) + (1 - \phi)\,p\,(q|u_{sam})\,, \tag{8}$$

where the interpolation weigh $\phi = S_{tail,sample} = sim(h_{tail}, h_{sample})$. This ensures that neighborhood enrichment reflects actual node relatedness and prevents noisy connections between dissimilar nodes. For each head node u_{head}, we define the similarity distribution for purification. Specifically, the similarity distribution for node u is defined as:

$$p_{sim}(q|u) = \begin{cases} sim(h_q, h_u), & q \in N(u) \\ 0, & q \notin N(u) \end{cases}, \tag{9}$$

where $sim(h_q, h_u)$ is the similarity between node q and node u. We sample $d_{head}(1 - p_{edrop})$ neighbors without replacement based on the similarity distribution $p_{sim}(q|u_{head})$, where p_{edrop} is the edge drop rate. This sampling removes the edges of dissimilar nodes, preserving effective neighborhood information.

By employing the feature augmentation and topology augmentation modules mentioned earlier, we create a new augmented modality graph from the original modality graph. This augmented modality graph is ultimately formulated as follows: $\hat{\mathcal{G}} = \{\hat{G}_1, \hat{G}_2, \cdots, \hat{G}_v\}$.

3.3. The Multi-modal Graph Contrastive Learning module

We input the original modality graph and the augmented modality graph into the encoder[4], resulting in node representations as $H = f(G_i)$ and $\hat{H} = f\left(\{\hat{G}\}_i\right)$ respectively.

We employ a contrastive objective [37], which aims for the encoded embeddings of each node from different modality graphs to align while distinguishable from embeddings of other nodes in the obtained node representations of two graphs. For any node u_i, its generated embedding in the original modality graph, named h_i, serves as the anchor. The subsequent embedding $\hat{h}_i$ in the augmented modality graph is treated as the positive sample, while embeddings of other nodes in the two modality graphs are naturally regarded as negative samples. Consequently, we define the pairwise objective $\ell(h_i, \hat{h}_i)$ for each positive pair $(h_i, \hat{h}_i)$ as:

$$\ell\left(h_i, \hat{h}_i\right) = -\log \frac{e^{\frac{\theta(h_i,\hat{h}_i)}{\tau}}}{e^{\frac{\theta(h_i,\hat{h}_i)}{\tau}} + \sum_{k\neq i} e^{\frac{\theta(h_i,\hat{h}_k)}{\tau}} + \sum_{k\neq i} e^{\frac{\theta(h_i,h_k)}{\tau}}}, \tag{10}$$

where τ is a temperature parameter. The $\theta(h, \hat{h})$ is defined as $sim(g(h), g(\hat{h}))$, where the projection $g(\cdot)$ is a two-layer multilayer perceptron (MLP) to enhance the expressive power [38]. $e^{\frac{\theta(h_i,\hat{h}_i)}{\tau}}$ is the positive pair, $\sum_{k\neq i} e^{\frac{\theta(h_i,\hat{h}_k)}{\tau}}$ is the inter-modality graph negative pairs, and $\sum_{k\neq i} e^{\frac{\theta(h_i,h_k)}{\tau}}$ is the intra-modality graph negative pairs. The overall objective to be maximized is the average of all positive pairs, formally as:

$$\mathcal{L}_{CL} = \frac{1}{2N} \sum_{i=1}^{N} \left[\ell(h_i, \hat{h}_i) + \ell(\hat{h}_i, h_i)\right], \tag{11}$$

where $\mathcal{L}_{CL}$ is the contrastive loss. Finally, integrate the multi-modal feature representation vector to maximize the multi-modal representation $Z = \{Z_1, Z_2, \cdots, Z_v\}$, thus learning the comprehensive modal representation vector. To mitigate the challenge of training with imbalanced data, the class-balanced loss incorporates a weighting factor that is inversely related to the effective number of samples. In our model, the output of the v-th modality is represented as $Z_v = [Z_{v_1}, Z_{v_2}, \cdots, Z_{v_K}]^T$. Subsequently, we calculate the probability distribution across all classes using the softmax function [39, 40], defined as follows:

$$p_{v_i} = \frac{\exp(Z_{v_i})}{\sum_{j=1}^{K} \exp(Z_{v_j})}. \tag{12}$$

Given a sample with class label y, the softmax cross-entropy loss for this sample is written as:

$$\mathcal{L}_{CB_{softmax}} = CB_{softmax}(Z_v, y) = -\frac{1-\beta}{1-\beta^{n_y}} \log\left(\frac{\exp\left(Z_{v_y}\right)}{\sum_{j=1}^{K} \exp\left(Z_{v_j}\right)}\right). \tag{13}$$

The overall loss function of DACL is computed as:

$$Loss = \mathcal{L}_{CL} + \mathcal{L}_{CB_{softmax}}. \tag{14}$$

3.4. The Multi-modal Integration module

To achieve a more comprehensive feature representation, we design a modality-level fusion self-attention mechanism that captures dynamic modality importance for each modality and facilitates adaptive cross-modal fusion from diverse modalities. The modality-level fusion self-attention mechanism calculates its attention value for each modality, denoting the attention value of the v-th modality as follows:

$$\varphi_v = Tanh\left(W \cdot (Z_v)^T + b\right), \tag{15}$$

where W is the weight matrix and b is the bias vector. After obtaining the self-attention value of the v-th modality, the Softmax activation function is introduced to learn the weights of different modalities as follows:

$$\beta_v = Softmax\left(\varphi_v\right) = \frac{\exp\left(\varphi_v\right)}{\exp\left(\varphi_1\right) + \exp\left(\varphi_2\right) + ... + \exp\left(\varphi_v\right)}, \tag{16}$$

where β_v is the weight of the v-th modality. After learning different modality weights, adding them to all modalities to get the expression of fusion, formally as:

$$Z_{mix} = \beta_1 Z_1{}^{(l)} \odot \beta_2 Z_2{}^{(l)} \odot \cdots \odot \beta_v Z_v{}^{(l)}, \tag{17}$$

where $\odot$ denotes the multiplication of the corresponding elements of each modality and $Z_v{}^{(l)}$ is the embedding feature of the v-th modality after l layers of convolution. Instead of using the previous linear summation approach to fuse information from different modalities, Z_{mix} uses the cross-multiplication of deep information from multiple modalities. The final fusion classification loss function of DACL is computed as:

$$\mathcal{L}_{fusion} = CB_{softmax}\left(Z_{mix}, y\right) = -\frac{1-\beta}{1-\beta^{n_y}} \log\left(\frac{\exp\left(Z_{mix_y}\right)}{\sum_{j=1}^{K} \exp\left(Z_{mix_j}\right)}\right), \tag{18}$$

and the specific algorithmic process is shown in Algorithm 1.

Algorithm 1 The DACL algorithm

Input: Multi-modal data $\mathcal{X} = \{X_1, X_2, \cdots, X_v\}$, adjacency matrices $\mathcal{A} = \{A_1, A_2, \cdots, A_v\}$ and semi-supervised information Y.

Stopping Criteria: Relative change in total loss $< 10^{-5}$ for 10 consecutive epochs or maximum 500 epochs.

Output: Classification result.

for $i = 1 \rightarrow v$ **do**
  Generate a augmented modality graph $\hat{G}_i$ for the i-th modality via topology augmentation and feature augmentation
end for
while not convergent **do**
  for $i = 1 \rightarrow v$ **do**
    Obtain node embeddings H of G_i using the encoder f
    Acquire node embeddings $\hat{H}$ of $\hat{G}_i$ through the encoder f
    Compute the contrastive loss $\mathcal{L}_{CL}$ with Equation (11), count the classification loss $\mathcal{L}_{CB_{softmax}}$ with Equation (13), and calculate the overall loss with Equation (14)
    Update parameters by applying stochastic gradient ascent to maximize $Loss$
  end for
  Get the multi-modal fusion representation Z_{mix} with Equation (15)- Equation (17)
  Compute the fusion classification loss $\mathcal{L}_{fusion}$ with Equation (18)
  Update parameters via stochastic gradient ascent to maximize $\mathcal{L}_{fusion}$
end while
return Classification result.

4. Experimental research

We evaluate eight methods on nine publicly available multi-modal datasets to assess the effectiveness of our proposed DACL approach. Our investigation is driven by four main research questions:

RQ1: Does the incorporation of GCN lead to enhanced classification performance? How does our method compare against existing state-of-the-art models designed for learning from multi-modal data?

RQ2: Does our dual graph augmentation method effectively mitigate the imbalance problem compared to other graph augmentation techniques?

RQ3: What impact does the inclusion of the dual graph augmentation module, the multi-modal graph contrastive learning module, and the multi-modal integration module have on the final classification performance?

RQ4: How do specific model hyperparameters, specifically the threshold and the edge drop rate, influence the ultimate performance of our model?

4.1. Experimental Settings

4.1.1. Datasets

The performance evaluation encompasses the following nine publicly available multi-modal datasets:

Animals consists of 30475 animal images categorized into 50 classes. We create a subset containing 10158 samples for our evaluation, selecting approximately 1/3 samples per class from the Animals, and extract two types of features from these images using DECAF and VGG19.

Caltech101[1] is a dataset containing 101 classes of images used for object recognition. It includes six modalities, each representing a different feature extracted from the original images: 48-D Gabor feature, 40-D wavelet moments (WM), 254-D CENTRIST feature, 1984-D HOG feature, 512-D GIST feature, and 928-D LBP feature.

Citeseer is a dataset comprising 3312 research paper samples classified into 6 categories. Following the setup described in [41], we create a 3703-D bag-of-words representation as the first modality and 3312-D vectors representing citation relationships between documents as the second modality.

GRAZ02[2] is an object categorization dataset comprising four image classes: bicycles, people, cars, and a class containing images without these objects. The dataset includes six extracted feature types: SIFT, SURF, GIST, LBP, PHOG, and WT.

NGs[3] is a subset of the 20 Newsgroup datasets, containing around 20000 newsgroup documents. It consists of 500 newsgroup documents classified into five classes. Each document is pre-processed using three different methods, resulting in three modalities of the data.

MNIST is a widely recognized dataset of handwritten digits. It includes three types of extracted features: 30-D IsoProjection, 9-D Linear Discriminant Analysis, and 9-D Neighborhood Preserving Embedding features.

Out-Scene is a dataset comprising 2688 images divided into eight groups. Four types of features are extracted for each image: 512-D GIST feature, 432-D color moment feature, 256-D HOG feature, and 48-D LBP feature.

[1] http://www.vision.caltech.edu/ImageDatasets/Caltech101/Caltech101.html
[2] http://www.emt.tugraz.at/~pinz/data/GRAZ_02
[3] http://www.qwone.com/~jason/20Newsgroups

NoisyMNIST[4] comprises 30000 images generated from the MNIST dataset by introducing white Gaussian noise, motion blur, and a combination of additive white Gaussian noise and reduced contrast.

Reuters[5] dataset contains 18758 documents categorized into six categories. Each document is written in five different languages: English, French, German, Italian, and Spanish.

Table 1 provides a statistical summary of these datasets, including the number of modalities, features, and classes for each dataset. The imbalanced ratio (IR) is defined as $\frac{Max}{Min}$, where the Max (Min) indicates the quantity of majority (minority) class examples.

Table 1: Detailed statistics of all datasets.

Datasets	Samples	Modalities	Features	Classes	Max	Min	IR
Animals	10158	2	4096 / 4096	50	389	31	12.55
Caltech101	9144	6	48 / 40 / 254 / 1984 / 512 / 928	101	800	31	25.81
Citeseer	3312	2	3703 / 3312	6	701	249	2.82
GRAZ02	1476	6	512 / 32 / 256 / 500 / 500 / 680	4	420	311	1.35
NGs	500	3	2000 / 2000 / 2000	5	100	100	1
MNIST	2000	3	30 / 9 / 9	10	327	69	4.74
Out-Scene	2688	4	512 / 432 / 256 / 45	8	410	260	1.58
NoisyMNIST	30000	2	784 / 784	10	3448	2720	1.27
Reuters	18758	5	21531 / 24892 / 34251 / 15506 / 11547	6	5102	1207	4.23

4.1.2. Evaluation indicators

To evaluate the performance of the models, we chose two indicators: overall classification accuracy (ACC) and macro-F1 score (F1). Specifically, we count the correct identification of positive labels in True Positives TP, incorrect identification of positive labels in False Positives FP and incorrect identification of negative labels in False Negative TN. ACC is the model's overall classification accuracy, the ratio of correctly classified samples to total samples. F1 is the precision and recall rate average across all classes. The above evaluation indicators are all positively associated with the models' performance. Then, we calculate the ACC and F1 as:

$$ACC = \frac{1}{N}\sum_{i=1}^{K} TP_i, \tag{19}$$

$$F1 = \frac{2}{K}\sum_{i=1}^{K} \frac{\mathrm{TP}_i}{2\mathrm{T}P_i + \mathrm{F}P_i + \mathrm{F}N_i}. \tag{20}$$

4.1.3. Parameter Settings

As for the proposed DACL, the framework uses a 2-layer GCN as the encoder. Each dataset's maximum number of iterations varies since we stop training when the loss converges. Training terminates when the relative change in total loss falls below 10^{-5} for 10 consecutive epochs, or after a maximum of 500 epochs to ensure computational feasibility. The Adam optimizer is applied to update the learnable parameters with the learning rate $lr = 0.01$. ReLu is chosen as the activation function. The size of hidden units is typically determined based on the feature dimension of the dataset, and it varies across different datasets due to their diverse feature dimensions. Our experiments are implemented by the PyTorch platform on a computer with an Intel i9-12900H CPU, Nvidia RTX 3060 GPU, and 16G RAM.

[4]https://github.com/nineleven/NoisyMNIST

[5]https://www.kaggle.com/datasets/nltkdata/reuter

4.2. Multi-modal Semi-supervised Classification(RQ1)

4.2.1. Compared Methods

We compare our DACL model with eight state-of-the-art methods to demonstrate its effectiveness. The baselines are divided into two categories: regular models that solely utilize features without considering graph convolutional network, and GCN-based models that incorporate graph convolutional network into the modelling process. Regular models include AMGL [41], MVAR [42], MLAN [23] and AWDR [24]. GCN-based models contain GCN Fusion [4], Co-GCN [25], DSRL [43] and IMvGCN [27]. The description of these methods and some detailed settings are given below.

AMGL is a parameter-free, self-weighted multi-graph learning algorithm.

MVAR is a scalable method that automatically balances the contributions of different modalities by adaptively optimizing the weight coefficients of each modality. We tune the trade-off weight for modality as $\lambda = 1000$, and the redistribution parameter γ is fixed as 2.

MLAN is an efficient algorithm that automatically assigns the desired weights to modality without additional weights and penalty parameters. The number of adaptive neighbors is tuned in [1, 10] to find the best performance on diverse datasets.

AWDR is a supervised multi-modal learning algorithm that assigns the best-learned weights to features from different modalities. The trade-off parameter λ is set as 1.0.

GCN Fusion is a method for processing multi-modal data based on the GCN model by constructing an average-weighted adjacency matrix before performing graph convolution. We utilize a 2-layer GCN in our experiments, with the learning rate at 0.01.

Co-GCN is a GCN-based method that exploits graph information from multiple modalities through the adaptive combination of graph Laplacian matrices. Note that the settings of the graph convolutional layers and learning rate are the same as those in GCN fusion.

DSRL is an end-to-end framework that constructs a neural network with multiple blocks consisting of learnable activation functions. The number of layers in the framework is fixed at 10.

IMvGCN is a method that combines reconstruction errors and Laplace embeddings to explore the original space from the perspective of features and topology and establish potential connections between GCN and multi-modal learning. We use a 3-layer neural network in our experiments, with a learning rate of 0.01.

4.2.2. Performance Comparison

We conducted extensive experiments five times in multi-modal node classification tasks, comparing DACL with the eight selected methods. The average performance of all compared approaches is summarized in Table 2, utilizing the classification accuracy (ACC) and F1 as evaluation metrics. It is important to note that all the compared approach results presented in our experiment are sourced from IMvGCN[27]. With bolded and underlined lines, we highlight the best and second-best findings. From the results, we have the following observations.

GCN improves the multi-modal representation. GCN-based methods consistently outperform regular models across a majority of datasets. This underscores the remarkable enhancement in feature representation capabilities achieved by integrating graph convolutional networks (GCNs) into the realm of multi-modal representation and integration. This highlights the inherent advantages of adopting graph-based approaches within the context of multi-modal learning.

For instance, our best-performing GCN-based model, DACL, surpasses the best regular model by notable margins. On the NGs dataset, our proposed DACL demonstrates improvements of 3.9% in ACC and 3.8% in F1, and similar enhancements of 2.1% and 2.2% are observed on the Reuters dataset. Moreover, when compared to the best regular model, DACL shines on Citeseer, with remarkable improvements of 8.4% in ACC and 7.7% in F1. Similarly, DACL showcases enhancements of 2.5% and 3.3% on the Animals dataset. Notably, these outcomes affirm that DACL stands out as the most proficient among all GCN-based models considered in the study.

This can be attributed to the innate capacity of GCNs to effectively capture intricate relationships and interactions among nodes, particularly when dealing with graph data. GCNs efficiently leverage the structural information inherent in graph data, facilitating the extraction of more comprehensive and enriched feature representations.

Our model is the best approach for effectively leveraging multi-modal information to extract features and addressing the challenges of multi-modal imbalance learning on most multi-modal datasets. The DACL model consistently delivers impressive performance across a variety of baseline methods, demonstrating strong outcomes across all evaluation metrics. Moreover, its performance remains robust even if it doesn't secure the top position in every dataset.

For instance, focusing on the ACC metric, the DACL model achieves a notable improvement of approximately 1.7% and 0.6% over the second-best values on the Out-Scene and Citeseer datasets, respectively. This enhancement can be attributed to the innovative multi-modal fusion self-attention mechanism introduced in our model. This mechanism dynamically adjusts the weightage assigned to modality, effectively capturing the features of minority-class nodes within modalities. This leads to a more balanced and accurate contribution of different modalities towards the final feature representation.

However, it's important to note that DACL's performance on the Caltech101 dataset fell short of surpassing that of IMvGCN, as initially anticipated. This could be attributed to the Caltech101 dataset's inherent multi-modal nature, which exhibits significant variations in dimensions. IMvGCN's integration of reconstruction error and Laplacian embedding contributes to a better preservation of the local invariance within each modality's manifold structure. In contrast, our model prioritizes addressing data imbalance concerns, which might have affected its performance in this scenario.

When examining the F1 metric, DACL consistently outperforms the second-best values by achieving an approximate improvement of 1.5% on the Animals dataset and 0.9% on the MNIST dataset. This progress can be accredited to our innovative dual-graph augmentation approach, designed to maintain the graph's balanced intrinsic structure and attributes. Additionally, incorporating a joint contrastive loss function significantly balances the weights assigned to different categories, effectively mitigating the challenges associated with node imbalance learning.

F1 score, which combines precision and recall, is a more appropriate evaluation metric for imbalanced datasets. Our model exhibits strong performance on highly imbalanced datasets. For instance, on the Animals dataset with an imbalance ratio of 12.55 and the MNIST dataset with a ratio of 4.74, our model achieves significant improvements of 1.5% and 0.9% in F1 score, respectively. This improvement underscores our model's effectiveness in handling highly imbalanced data by mitigating the impact of redundant information, balancing the topology between nodes, and ensuring appropriate attention to each class during the learning process without bias toward classes with more samples.

4.3. Comparison of graph augmentation performance (RQ2)

To demonstrate the effectiveness of our dual graph augmentation, we compare it with seven state-of-the-art graph augmentations. These graph augmentations include diffusion with Per-

Table 2: Classification ACC and F1 (mean% and standard deviation%) of all compared multi-modal semi-supervised classification methods with 5% labeled samples as supervision, where the best performance is highlighted in bold and the second best result is underlined. AMGL fails to work on some datasets, which are marked with ”—” in the table, and OOM denotes the out-of-memory error.

Category	Methods	Metrics	Animals	Citeseer	GRAZ02	NGs	Out-Scene	MNIST	Caltech101	NoisyMNIST	Reuters
Regular	AMGL	ACC	66.3±0.7	——	50.9±1.5	——	64.2±1.5	65.9±2.3	42.0±0.5	93.9±1.1	——
		F1	60.5±0.8	——	51.7±1.6	——	66.0±1.4	67.1±1.9	25.1±0.7	93.9±1.1	——
	MVAR	ACC	78.0±1.1	63.1±1.2	49.0±3.8	29.1±9.1	42.5±1.5	78.2±3.9	44.3±1.0	73.3±0.8	82.3±0.0
		F1	73.0±1.1	59.2±1.9	50.5±3.6	40.5±9.4	44.5±1.9	76.3±4.7	25.1±1.4	72.7±0.7	80.0±0.0
	MLAN	ACC	78.8±0.7	62.3±4.5	50.2±6.3	93.8±1.1	69.0±2.0	85.6±1.2	35.6±0.1	OOM	OOM
		F1	72.9±1.0	59.2±2.3	52.4±2.7	93.9±1.1	72.5±1.6	82.7±1.4	16.9±0.7	OOM	OOM
	AWDR	ACC	79.3±0.7	62.0±1.4	40.9±1.2	54.1±5.4	52.9±1.6	66.4±4.8	37.2±0.7	76.2±0.5	OOM
		F1	72.6±0.9	57.5±0.9	40.5±1.5	62.9±5.7	54.3±1.4	65.3±3.6	16.6±0.7	75.9±0.5	OOM
	DSRL	ACC	73.8±0.7	58.8±1.8	46.9±3.5	71.4±2.8	61.9±2.0	86.7±1.2	48.8±0.6	OOM	OOM
		F1	67.0±1.7	53.8±1.7	47.1±3.7	71.7±2.8	62.5±2.0	83.0±3.1	27.8±1.2	OOM	OOM
GCN-based	GCN Fusion	ACC	75.8±0.7	60.0±1.1	53.1±2.6	92.7±1.4	69.6±1.2	83.5±1.2	47.8±0.7	90.2±1.8	81.2±0.4
		F1	67.7±1.0	52.4±0.5	53.6±2.7	92.7±1.4	69.9±1.4	81.1±1.2	27.4±0.5	90.1±1.5	79.0±0.4
	Co-GCN	ACC	67.7±1.0	52.4±0.5	53.6±2.7	92.7±1.4	69.9±1.4	81.1±1.2	27.4±0.5	90.1±1.5	79.0±0.4
		F1	67.7±1.0	52.4±0.5	53.6±2.7	92.7±1.4	69.9±1.4	81.1±1.2	27.4±0.5	90.1±1.5	79.0±0.4
	IMvGCN	ACC	81.6±0.1	70.9±0.1	58.6±0.6	97.5±0.1	74.3±0.3	90.8±0.3	**51.7±0.2**	94.6±0.4	84.2±0.0
		F1	74.8±0.2	65.6±0.3	58.7±0.7	97.5±0.1	74.5±0.3	88.6±0.2	30.7±0.2	94.4±0.4	81.7±0.0
	DACL	ACC	**81.8±0.2**	**71.5±0.3**	**59.1±0.2**	**97.7±0.2**	**76.0±0.8**	**91.1±0.1**	50.9±0.2	**94.8±0.1**	**84.4±0.1**
		F1	**76.3±0.3**	**66.9±0.3**	**59.6±0.5**	**97.7±0.2**	**76.3±0.3**	**89.5±0.2**	**30.9±0.3**	**94.7±0.1**	**82.2±0.1**

sonalized PageRank (PPR)[31], diffusion with Markov Diffusion Kernels (MDK)[44], Edge Removing and Feature Masking (ERFM)[45], Edge Removing and Feature Dropout (ERFD)[46], Edge Removing and Feature Masking with Eigenvector (ERFM-EV)[32], Edge Removing and Feature Masking with Degree (ERFM-DE)[32] and Edge Removing and Feature Masking with PageRank (ERFM-PR)[32]. The classification ACC and F1 are shown in Table 3. Based on the results obtained, the following observations can be made.

The distinct augmentation strategies exhibit strengths on different datasets, while DGA demonstrates a substantial advantage over the compared algorithms across all datasets. For example, Out-Scene shows a 1% and 0.9% enhancement in F1 value compared to the ERFM and ERFD, respectively. GRAZ02 achieves a 0.8% and 0.8% increment in F1 compared to the PPR and MDK, respectively. Moreover, our DGA outperforms the best baseline of ERFM-EV, ERFM-DE and ERFM-PR by 0.8% in ACC and 1.1% in F1 on the MNIST dataset.

Table 3: Classification ACC and F1 (mean% and standard deviation%) of all compared graph augmentation methods with 5% labeled samples as supervision.

Methods	Metrics	Animals	Citeseer	GRAZ02	NGs	Out-Scene	MNIST	Caltech101	NoisyMNIST	Reuters
ERFM	ACC	81.3±0.7	70.7±0.7	58.4±1.0	94.4±1.9	75.2±0.6	90.1±0.5	50.2±0.2	94.2±0.4	84.3±0.1
	F1	75.7±0.9	66.3±0.4	58.3±1.3	94.4±1.9	75.3±0.5	87.9±0.9	29.9±0.4	94.1±0.4	82.1±0.2
ERFD	ACC	81.7±0.6	70.2±0.7	58.6±0.5	92.5±3.0	75.3±0.3	89.0±1.1	50.3±0.3	94.4±0.2	84.2±0.1
	F1	76.2±0.7	66.0±0.4	58.7±0.8	92.5±3.0	75.4±0.3	87.1±1.1	29.9±0.4	94.3±0.2	82.0±0.1
PPR	ACC	81.5±0.5	68.2±0.9	58.6±0.8	95.0±0.7	75.8±0.3	90.2±0.5	50.1±0.2	94.4±0.4	84.3±0.2
	F1	75.8±0.6	64.4±1.0	58.8±1.1	95.0±0.7	75.9±0.3	88.3±0.4	29.7±0.3	94.3±0.4	82.1±0.2
MDK	ACC	81.5±0.4	68.2±0.9	58.6±0.8	95.0±0.7	75.9±0.2	90.2±0.5	50.1±0.2	94.4±0.4	84.3±0.2
	F1	75.9±0.8	64.4±1.0	58.8±1.1	95.0±0.7	76.1±0.3	88.4±0.3	29.7±0.3	94.3±0.4	82.1±0.2
ERFM-EV	ACC	81.5±0.6	69.3±1.0	57.7±1.4	95.2±1.5	75.9±0.5	90.3±0.5	50.1±0.2	94.2±0.4	84.3±0.1
	F1	75.9±0.7	65.3±0.6	57.7±1.9	95.2±1.5	76.0±0.5	88.4±0.4	29.7±0.2	94.1±0.4	82.1±0.2
ERFM-DE	ACC	81.5±0.6	69.3±1.0	57.7±1.4	95.2±1.5	75.9±0.6	90.3±0.5	50.1±0.2	94.2±0.4	84.3±0.1
	F1	75.9±0.7	65.3±0.6	57.7±1.9	95.2±1.5	76.1±0.6	88.4±0.4	29.7±0.2	94.1±0.4	82.1±0.2
ERFM-PR	ACC	81.5±0.6	69.3±1.0	57.7±1.4	95.2±1.5	75.9±0.6	90.3±0.5	50.1±0.2	94.2±0.4	84.3±0.1
	F1	75.9±0.7	65.3±0.6	57.7±1.9	95.2±1.5	76.1±0.6	88.4±0.4	29.7±0.2	94.1±0.4	82.1±0.2
DGA	ACC	**81.8±0.2**	**71.5±0.3**	**59.1±0.2**	**97.7±0.2**	**76.0±0.8**	**91.1±0.1**	**50.9±0.2**	**94.8±0.1**	**84.4±0.1**
	F1	**76.3±0.3**	**66.9±0.3**	**59.6±0.5**	**97.7±0.2**	**76.3±0.3**	**89.5±0.2**	**30.9±0.3**	**94.7±0.1**	**82.2±0.1**

This can be attributed to two key factors. Firstly, in contrast to other augmentation approaches, our DGA employs a differential topology augmentation strategy for the head and tail nodes within the graph structure. Specifically, we apply a self-centring network interpolation method to enrich the neighborhood information of tail nodes, while for head nodes, we perform similarity-based sampling within their neighborhoods to narrow the scope of their surroundings. This tailored approach to adjusting the topology relationships among nodes contributes to a more balanced graph structure, effectively mitigating the impact of topology imbalance. Secondly, in our attribute enhancement strategy, DGA computes the PageRank centrality of nodes during the feature masking process. This technique aids in preserving the critical feature dimensions of nodes and uncovers a more comprehensive underlying feature representation. This dual focus on topology and attribute enhancement within our DGA method collectively contributes to its enhanced performance.

4.4. Ablation study analysis (RQ3)

To illustrate the advantages of MmI, MmGCL, and DGA components in our model, we conduct ablation experiments using the Citeseer dataset, for example. Table 4 records the experimental results of the ablation study. Observing the results, it becomes evident that MmI, MmGCL, and DGA components enhance model performance.

When comparing models that do not utilize MmI, MmGCL, and DGA, the model incorporating MmI demonstrates improvements of 0.8% in ACC and 1.7% in F1, respectively. This enhancement can be attributed to MmI's ability to harness complementary information from various modalities, discern the unique contributions of each modality to the classification task, and enhance the comprehensiveness of feature representation.

Furthermore, it's evident that the model equipped with both MmI and MmGCL outperforms the model without MmGCL by 0.1% in ACC and 4.6% in F1, respectively. This underscores the effectiveness of the joint contrastive loss function in MmGCL, which promotes the consistency of node features across diverse modalities, thereby bolstering the model's capability to classify nodes from minority classes. Consequently, this enhancement significantly improves the model's evaluation metrics.

Table 4: Ablation study (mean% and standard deviation%) of the proposed method on the Citeseer dataset, where the experiment is conducted with 5% labelled samples.

MmI	MmGCL	DGA	ACC	F1
			69.7±0.7	60.4±0.5
✓			70.5±0.4	62.1±1.0
✓	✓		70.6±0.3	66.7±0.4
✓	✓	✓	**71.5±0.3**	**66.9±0.3**

In addition, upon closer examination, the model integrating MmI, MmGCL, and DGA showed enhancements of 0.9% in ACC and 0.2% in F1. This improvement can be attributed to the dual augmentation strategies employed. These strategies assist in reducing potential noise and simultaneously adjust the neighborhoods of different nodes. Consequently, this adjustment leads to a more balanced distribution of information within the topological structure, effectively mitigating the issue of topology imbalance across modalities.

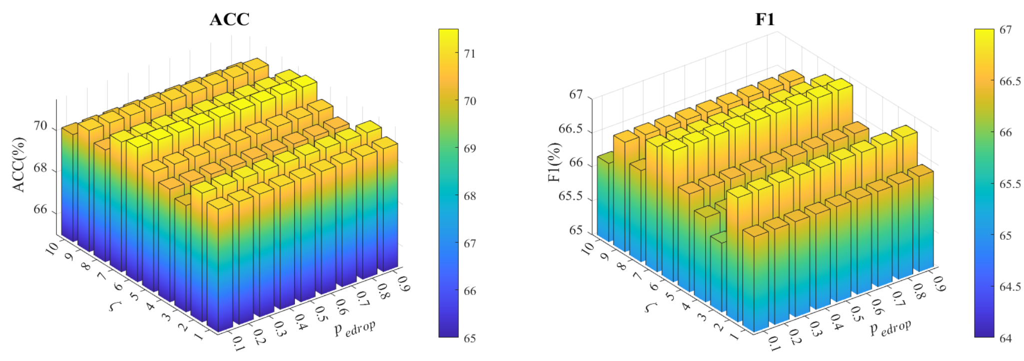
4.5. Sensitivity analysis (RQ4)

To demonstrate the stability of the model under the perturbation of these crucial hyperparameters, we undertake a sensitivity analysis on the two hyperparameters ζ and p_{edrop} that influence the generation of graph modalities. Figure 2 shows the impacts of ζ and p_{edrop} on the Citeseer dataset , for instance.

The results highlight that the selection of ζ notably impacts classification performance, whereas the choice of p_{edrop} does not influence classification performance. Specifically, when ζ is increased from 1 to 6, both ACC and F1 metrics consistently improve. However, when ζ is further increased from 6 to 10, both ACC and F1 exhibit a declining trend. This behaviour suggests that an optimal value for ζ exists that balances between addressing tail node neighbor sparsity and avoiding noise introduction.

Moreover, as ζ increases from 1 to 10, no significant changes in model performance are observed. This indicates that the optimal discrimination across head and tail nodes is achieved at around ζ value of 6. Additionally, the hyperparameter p_{edrop} can be chosen from the range of [0.1, 0.9], as it doesn't substantially impact model performance.

In summary, our DACL model demonstrates promising performance with well-suited hyperparameters. Furthermore, DACL consistently maintains strong performance across a broad spectrum of parameter settings, underscoring its robustness and effectiveness.



(a) ACC (b) F1

Figure 2: Parameter sensitivity demonstration of the proposed method on the Citeseer dataset with average ACC and F1.

5. Conclusion

In this study, we present DACL, an integrated framework that addresses imbalanced multi-modal representation through dual graph augmentation, contrastive learning, and cross-modal integration. By combining feature-aware pruning to eliminate noisy attributes with modality-specific augmentation strategies that rebalance node degree distributions, our approach effectively mitigates both topological and quantitative imbalances. The contrastive learning module enhances minority-class representation through targeted sample alignment, while adaptive modality fusion dynamically combines cross-modal features to improve tail-class discrimination. Extensive experiments demonstrate DACL's superior performance in generating balanced node representations across diverse imbalance scenarios. This work advances imbalanced learning in graph-structured multi-modal systems by providing a principled solution that harmonizes structural and feature-level imbalances while preserving critical semantic relationships. While our method demonstrates robust performance for datasets at the ten-thousand-sample scale, its computational efficiency requires further optimization when applied to extreme-scale graphs. The similarity computation, while effective for current experimental scales, may face scalability challenges when processing orders-of-magnitude larger graphs. Future research directions could explore approximate similarity measures or distributed computing strategies to maintain the method's advantages while enhancing its computational efficiency for ultra-large-scale applications.

Author Contributions

Shuang Yao (yaoshuang@hebut.edu.cn): responsible for conceptualization, investigation, writing-original draft, review & editing. Wenfeng Han (202432805008@stu.hebut .edu.cn): responsible for data curation, software, validation, formal analysis, writing-original draft. Xinxin Wei (m13650156018@163.com): responsible for resources, software, validation, formal analysis. Yao Dong (dongyao@hebut.edu.cn): responsible for methodology, funding acquisition, writing-review & editing. Tianyi Zhang (202432803056@stu.hebut.edu.cn): responsible for data curation, visualization. Yongfeng Dong (dongyf@hebut.edu.cn): responsible for supervision, project administration, writing-review & editing.

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Author Biography

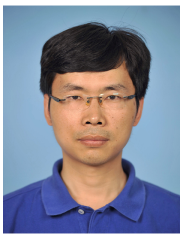
Shuang Yao received the Master of Engineering degree from the School of Computer Science and Engineering, Hebei University of Technology, Tianjin, China, in 2014. She is currently an Assistant Researcher with the School of Artificial Intelligence, Hebei University of Technology. Her main research areas are multi-modal learning, data mining.

Wenfeng Han received the degree of bachelor in software engineering from North China Institute of Aerospace Engineering, China, in 2024. He is currently working toward the M.S degree in Hebei University of Technology. His research focuses on imbalanced multi-modal learning.

Xinxin Wei graduated with a bachelor's degree in Information Security from Hebei Normal University, Hebei, China, in 2021. She graduated with a master's degree in Computing Technology from Hebei University of Technology in 2024.Her research focuses on machine learning and multi-modal learning.

Yao Dong received the B.S. and M.S. degrees in electronic information engineer and communication and information system at Hebei University of Technology, Tianjin, China, in 2005 and 2008. She is currently a Senior Experimentalist with the School of Artificial Intelligence, Hebei University of Technology. She is a member of the China Computer Federation. Her interests mainly include machine learning and artificial intelligence and multi-modal learning.

Tianyi Zhang received the degree of bachelor in computer science and technology from North China University of Science and Technology, China, in 2024. He is currently working toward the M.S degree in Hebei University of Technology. His research focuses on imbalanced data mining.



Yongfeng Dong is a Professor with the School of Artificial Intelligence. He is a Senior Member of the China Computer Federation (CCF), a member of the Education Committee of the CCF, the Standing Director of the Hebei Computer Federation, and the Director of the Hebei Institute of Electronics. His research interests include intelligent information processing and big data technology.